Chapter 5 Critical Thinking

Questions 1–9 and 11 • Textbook page 118

1. Purpose of a loop
A loop repeats a group of statements while a condition is true. It saves writing the same code over and over.

2. while and do-while
A while loop checks its condition before the code runs, so it can run zero times. A do-while loop checks after the code runs, so it always runs at least once.

3. Input validation
The Prompter review used an input validation loop. It kept asking until the number was between the minimum and maximum.

4. Infinite loops and overflow
a) An infinite loop keeps running because its condition never becomes false.
b) A syntax mistake, such as an extra semicolon or missing braces, can cause one. A logic mistake, such as changing the control variable in the wrong direction, can also cause one.
c) Overflow happens when a number becomes too large for the number of bits available to store it. The value can wrap around and change sign.

5. Number of iterations
The loop runs 60 times. x starts at 0 and increases by 2 until it reaches 120.

6. Infinite do-while loop
An initial value of 120 would make the loop keep going because 3 is subtracted each time and x stays below 120.

7. Counters and accumulators
Both are variables updated inside a loop. A counter changes by a fixed amount. It can count inputs or guesses. An accumulator adds changing values. It can total scores or add values before finding an average.

8. for statement
```
int sum = 0;
for (int i = 3; i <= 10; i++)
{
   sum += i;
}
```

9. Choosing a loop
I would check whether the number of repetitions is known and whether the code must run at least once. A for loop works well for a fixed number of repetitions, while do-while works when the body must run before the condition is checked.

11. String methods
For String x = "my string.";

a) x.length() returns 10
b) x.substring(0, 3) returns "my "
c) x.toLowerCase() returns "my string."
d) x.toUpperCase() returns "MY STRING."
e) x.trim() returns "my string."